

Statistics for Economists II

Lecture 1: Introduction to Statistical theories and Distributions

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Topic 1: Statistical Theories

Theory 1: Central Limit Theorem

Objectives: Upon completion of this lesson, you should be able to:

- To learn the Central Limit Theorem.
- To get an intuitive feeling for the Central Limit Theorem.
- To use the Central Limit Theorem to find probabilities concerning the sample mean.
- To be able to apply the methods learned in this lesson to new problems.

Theorem 1: Central Limit Theorem

- The Central Limit Theorem (CLT) is one of the most important results in statistics. It explains why the normal distribution is so widely used in statistical inference.
- It connects probability theory with real-world data.
- It allows us to make inferences about populations using sample statistics.

- If $X_1, X_2, \dots, X_n$ is a random sample from a normal distribution with mean μ and finite variance σ^2 , the random variable

$$\frac{\sum_1^n X_i - n\mu}{\sigma\sqrt{n}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$$

is for every positive integer n , normally distributed with zero mean and unit variance.

- A special case: If $X_1, X_2, \dots, X_n$ is a random sample of size n from a any distribution having finite variance $\sigma^2 > 0$ and a finite mean μ , then the random variable $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ converges in distribution to a random variable having a standard normal distribution, with mean zero and variance 1.

Theorem 1.1

Theorem 1.1. Let $X_1, X_2, \dots, X_n$ denote the observations of a random sample from a distribution that has mean μ and positive variance σ^2 . Then the random variable $Y_n = (\sum_1^n X_i - n\mu)/\sqrt{n}\sigma = \sqrt{n}(\bar{X} - \mu)/\sigma$ converges in distribution to a random variable which has a normal distribution with mean zero and variance 1.

4 Intuition Behind CLT

The CLT works because:

- Each observation contributes a small, independent effect.
- Aggregating many such effects leads to a bell-shaped (normal) distribution.
- Random fluctuations cancel out, leaving a stable pattern.

Even if the original population is: Skewed, Uniform, Exponential: the distribution of the sample mean becomes approximately normal for large n

Theorem 2: Markov's Inequality

- Markov's Inequality is a fundamental result in probability theory that provides an upper bound on the probability that a non-negative random variable takes on values much larger than its expected value. It is often used as a starting point for proving more refined inequalities, such as Chebyshev's inequality and the Weak Law of Large Numbers.
- **Theorem 1.2:** Let X be a non-negative random variable with finite expected value $E[X]$. Then, for any $a > 0$:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

- Intuition
 - i. If the mean of a random variable is small, the chance of it taking very large values must also be small.
 - ii. For example, if the average value of X is 10, the probability that X exceeds 100 cannot be more than $10/100 = 0.1$

Theorem 3: Chebyshev's inequality

- Chebyshev's inequality is one of the most important results in probability and statistics. It improves upon Markov's inequality by using not just the mean, but also the variance of a random variable, to bound how far the variable can deviate from its mean.

Theorem 1.3: Let X be a random variable with finite mean $\mu = E[X]$ and finite variance $\sigma^2 = \text{Var}[X]$. Then, for any $k > 0$:

$$P(|X - \mu| \geq k\sigma) = \frac{1}{k^2}$$

Equivalently:

$$P(|X - \mu| < k\sigma) = 1 - \frac{1}{k^2}$$

Interpretation of Chebyshev's inequality

- The probability that X lies more than k standard deviations away from the mean is at most $1/k^2$.
- This is distribution-free: it holds for all random variables with finite variance, regardless of shape (not just normal distributions).
- For example: At least 75% of observations lie within 2 standard deviations of the mean ($1 - 1/2^2 = 0.75$). At least 89% of observations lie within 3 standard deviations of the mean ($1 - 1/3^2 = 0.89$).

Limitations of Chebyshev's inequality

- Like Markov's inequality, Chebyshev's bound can be conservative (loose). For bell-shaped distributions like the normal, the actual probabilities are much smaller.
- However, its strength is universality: it works for any distribution with finite variance.

Theorem 4: Cauchy–Schwarz Inequality

- The Cauchy–Schwarz inequality is a fundamental result in linear algebra, analysis, and probability theory. It gives a bound on the inner product of two vectors (or random variables) in terms of their lengths (norms). It is widely used in proving other inequalities and results, such as the triangle inequality and correlation bounds.
- **Theorem 1.4** For any vectors u, v in an inner product space, the Cauchy–Schwarz inequality states:

$$|\langle u, v \rangle| \leq \|u\| \|v\|$$

where

- $\langle u, v \rangle$ is the inner product of u and v ,
- $\|u\| = \sqrt{\langle u, u \rangle}$ is the norm (length) of u .

Coordinate Form (Euclidean Space)

For vectors $u = (u_1, u_2, \dots, u_n)$ and $v = (v_1, v_2, \dots, v_n)$:

$$\left| \sum_{i=1}^n u_i v_i \right| \leq \sqrt{\sum_{i=1}^n u_i^2} \sqrt{\sum_{i=1}^n v_i^2}$$

In Probability and Statistics

For random variables X and Y with finite second moments:

$$|\mathbb{E}[XY]| \leq \sqrt{\mathbb{E}[X^2]} \sqrt{\mathbb{E}[Y^2]}$$

This form is used in correlation and covariance analysis.

Proof Outline

Algebraic Proof (Finite Dimensions)

- 1 Consider the quadratic function:

$$f(t) = \|u - tv\|^2 = \langle u - tv, u - tv \rangle \geq 0$$

for all $t \in \mathbb{R}$.

- 2 Expanding: $f(t) = \langle u, u \rangle - 2t\langle u, v \rangle + t^2\langle v, v \rangle$
- 3 Since $f(t) \geq 0$ for all t , the discriminant must be non-positive:

$$(-2\langle u, v \rangle)^2 - 4\langle u, u \rangle\langle v, v \rangle \leq 0$$

- 4 Simplifying: $\langle u, v \rangle^2 \leq \langle u, u \rangle\langle v, v \rangle$

Taking square roots gives: $|\langle u, v \rangle| \leq \|u\| \|v\|$

Equality Condition

Equality holds **if and only if** u and v are linearly dependent, i.e., $u = \lambda v$ for some scalar λ .

Applications

In Linear Algebra

- Used to prove triangle inequality for norms.
- Forms the basis of orthogonality concepts.

In Probability and Statistics

- Ensures the **correlation coefficient** satisfies:

$$-1 \leq \rho_{XY} \leq 1$$

- Provides bounds for covariance:

$$|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}$$

Examples

Example 1 (Vectors)

Let $u = (1, 2, 3)$, $v = (4, -5, 6)$.

- Inner product: $\langle u, v \rangle = 1(4) + 2(-5) + 3(6) = 12$.
- Norms: $\|u\| = \sqrt{14}$, $\|v\| = \sqrt{77}$.
- Inequality:

$$|12| \leq \sqrt{14}\sqrt{77} \approx 32.84$$

Verified.

Example 2 (Random Variables)

Let X and Y be random variables with $\mathbb{E}[X^2] = 9$, $\mathbb{E}[Y^2] = 4$, and $\mathbb{E}[XY] = 5$.

By Cauchy–Schwarz:

$$|5| \leq \sqrt{9}\sqrt{4} = 6$$

Holds true.

Theorem 5: Empirical Rule

The **Empirical Rule** (also known as the **68–95–99.7 Rule**) is a guideline in statistics that describes how data are distributed in a **normal distribution**. It provides approximate proportions of observations that fall within one, two, and three standard deviations of the mean. This rule is particularly useful for interpreting data, detecting outliers, and understanding variability.

Normal Distribution and the Empirical Rule

Normal Distribution

- The normal distribution is a bell-shaped, symmetric distribution defined by its mean (μ) and standard deviation (σ).
- Many natural and social phenomena (e.g., test scores, heights, measurement errors) approximate a normal distribution.

Theorem 1.4: The Empirical Rule

For a variable $X \sim N(\mu, \sigma^2)$:

- **68% Rule:** About 68% of observations fall within one standard deviation of the mean:

$$P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68$$

- **95% Rule:** About 95% of observations fall within two standard deviations:

$$P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$$

- **99.7% Rule:** About 99.7% of observations fall within three standard deviations:

$$P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.997$$

This is why the rule is often referred to as the **68 – 95 – 99.7 Rule**.

Data Spread

- The rule gives a **quick way to estimate variability** in data.
- It helps assess whether a dataset is approximately normal.

Outlier Detection

- Observations beyond 3σ are extremely rare ($< 0.3\%$), hence often considered **outliers**.

Comparisons

- The rule only holds for **normal (bell-shaped, symmetric) distributions**.

Visual Representation

Imagine a bell-shaped curve centered at the mean (μ):

- The interval $[\mu - \sigma, \mu + \sigma]$ covers 68% of the area under the curve.
- The interval $[\mu - 2\sigma, \mu + 2\sigma]$ covers 95%.
- The interval $[\mu - 3\sigma, \mu + 3\sigma]$ covers 99.7%.

This visualization helps students **see the proportions** of data spread.

Worked Examples

Example 1: Exam Scores

Suppose exam scores are normally distributed with mean $\mu = 70$ and standard deviation $\sigma = 10$.

- **Within** 1σ : $70 \pm 10 = [60, 80]$. About 68% of students scored between 60 and 80.
- **Within** 2σ : $70 \pm 20 = [50, 90]$. About 95% of students scored between 50 and 90.
- **Within** 3σ : $70 \pm 30 = [40, 100]$. About 99.7% of students scored between 40 and 100.

Example 2: Heights

If the average male height is 175 cm with $\sigma = 7$ cm:

- About 68% of men are between 168 cm and 182 cm.
- About 95% are between 161 cm and 189 cm.
- About 99.7% are between 154 cm and 196 cm.

Relation to Chebyshev's Theorem

- **Chebyshev's theorem** provides a general inequality for *any* distribution:

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}$$

- Unlike the Empirical Rule, it does not assume normality.
- For example, Chebyshev guarantees at least 75% of observations within 2σ , while the Empirical Rule states 95% if the data are normal.
- Thus, the Empirical Rule is **more precise but less general**.

Applications in Statistics

- **Quality control:** Detecting unusual production outcomes.
- **Finance:** Evaluating volatility of returns.
- **Education:** Assessing test scores against average performance.
- **Research:** Screening for outliers in experimental data.

Point Estimators and Their Properties

In inferential statistics, one of the main goals is to use sample data to make conclusions about unknown **population parameters** (e.g., mean, variance, proportion).

A **point estimator** is a statistic (a function of the sample data) that provides a **single best guess** (a point estimate) of an unknown parameter. For example:

- Sample mean ($\bar{X}$) estimates the population mean (μ).
- Sample variance (s^2) estimates the population variance (σ^2).
- Sample proportion ($\hat{p}$) estimates the population proportion (p).

The quality of an estimator is judged based on its **statistical properties**.

Definitions

- **Parameter:** A numerical value describing a population characteristic (fixed but unknown).
- **Estimator:** A rule or formula for calculating an estimate of the parameter based on sample data.
- **Estimate:** The actual numerical value computed from a sample using the estimator.

Example:

- Population mean: μ (unknown).
- Estimator: $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.
- Estimate: $\bar{x}$ (calculated value from a sample).

Desirable Properties of Point Estimators

- 1 Unbiasedness
- 2 Consistency
- 3 Efficiency
- 4 Sufficiency

Property No 1: Unbiasedness

- A fundamental property of an estimator is unbiasedness, which ensures the estimator targets the true parameter value on average.
- Definition of Unbiasedness:

$$E[\hat{\theta}] = \theta$$

- If the expected value of the estimator equals the true parameter, then $\hat{\theta}$ is called an unbiased estimator of θ .
- **Bias:** Bias is defined as

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

If $\text{Bias}(\hat{\theta}) = 0$, the estimator is **unbiased**.

Properties of Unbiased Estimators

- ① Linearity: If $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased estimators of θ_1 and θ_2 , then for constants a, b :

$$a\hat{\theta}_1 + b\hat{\theta}_2$$

is an unbiased estimator of $a\theta_1 + b\theta_2$

- ② Not Unique: There can be multiple unbiased estimators for the same parameter.
- ③ Efficiency: Among unbiased estimators, some are preferred because they have lower variance (leading to minimum variance unbiased estimators, MVUE).

Proofs of Unbiasedness for Common Estimators

Proof 1: Sample Mean is Unbiased

Let $X_1, X_2, \dots, X_n$ be i.i.d. with $E[X_i] = \mu$.

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Proof:

$$E[\bar{X}] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n}(n\mu) = \mu$$

- $\bar{X}$ is an unbiased estimator of μ .

Proof 2: Sample Variance is Unbiased

- Let $x_1, \dots, X_n$ be a random sample from a population with mean μ and variance σ^2 .
- Population variance: $\sigma^2 = E[(X - \mu)^2]$
- Estimator: $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$
- We want to prove that the estimator s^2 is unbiased estimator for σ^2 . That means we must show that

$$E[s^2] = \sigma^2$$

Proof: We start by expanding the summation component of the sample variance:

$$\begin{aligned}\sum (X - \bar{X})^2 &= \sum (X^2 - 2X\bar{X} - \bar{X}^2) \\ &= \sum X^2 - 2\bar{X} \sum X - \sum \bar{X}^2\end{aligned}$$

remember $\sum X = n\bar{X}$. and sum of a constant is a constant times n. Eg. $\sum \bar{X} = n\bar{X}$ Therefore,

$$\begin{aligned}&= \sum X^2 - 2n\bar{X}^2 - n\bar{X}^2 \\ &= \sum X^2 - n\bar{X}^2\end{aligned}$$

Therefore,

$$\begin{aligned} E[s^2] &= E\left[\frac{1}{n-1} \left(\sum X^2 - n\bar{X}^2\right)\right] \\ &= \frac{1}{n-1} \left(\sum E[X^2] - nE[\bar{X}^2]\right) \end{aligned}$$

Note that by definition: $E[X^2] = \sigma^2 + \mu^2$ and $E[\bar{X}^2] = \frac{\sigma^2}{n} + \mu^2$. so

$$\begin{aligned} &= \frac{1}{n-1} \left(\sum(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right)\right) \\ &= \frac{1}{n-1} \left(n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right)\right) \\ &= \frac{1}{n-1} \left(n\sigma^2 + n\mu^2 - \sigma^2 - n\mu^2\right) \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{n-1} (n\sigma^2 - \sigma^2) \\
 &= \frac{\sigma^2}{n-1} (n-1) \\
 &= \sigma^2
 \end{aligned}$$

Thus: $E[s^2] = \frac{\sigma^2}{n-1}(n-1) = \sigma^2$

- s^2 is unbiased for σ^2 .

Proof 3: MLE Variance is Biased

MLE for variance:

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

From previous result:

$$E \left[\sum_{i=1}^n (X_i - \bar{X})^2 \right] = (n-1)\sigma^2$$

$$E[\hat{\sigma}_{MLE}^2] = \frac{1}{n}(n-1)\sigma^2 = \frac{n-1}{n}\sigma^2$$

- MLE variance is **biased** but asymptotically unbiased.

Theorem: Linearity of Expectation

If $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased for θ_1 and θ_2 ,
then for constants a, b :

$$a\hat{\theta}_1 + b\hat{\theta}_2$$

is unbiased for $a\theta_1 + b\theta_2$.

Proof:

$$E[a\hat{\theta}_1 + b\hat{\theta}_2] = aE[\hat{\theta}_1] + bE[\hat{\theta}_2] = a\theta_1 + b\theta_2$$

Bias–Variance Decomposition

The **Mean Squared Error** (MSE):

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

Expand:

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + (\text{Bias}(\hat{\theta}))^2$$

- Shows the trade-off between variance and bias.

Summary

- Sample mean is unbiased for μ .
- Sample variance (with $n - 1$ denominator) is unbiased for σ^2 .
- MLE variance estimator is biased but asymptotically unbiased.
- Linearity preserves unbiasedness.
- MSE decomposes into variance and squared bias.

Property No 2: Consistency

- Estimators should not only be unbiased in finite samples but also behave well as the sample size grows. This property is called **consistency**.
- It ensures that with enough data, the estimator converges to the true parameter value.

Definition of Consistency

- Let $\hat{\theta}_n$ be an estimator of θ , based on a sample of size n :

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \text{as } n \rightarrow \infty$$

- Meaning for every $\epsilon > 0$:

$$\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| > \epsilon) = 0$$

- An estimator with this property is called **consistent**.

Types of Consistency

1 Weak Consistency

Let $\hat{\theta}_n$ be an estimator of θ , based on a sample of size n .

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \text{as } n \rightarrow \infty$$

Meaning:

$$\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| > \epsilon) = 0, \quad \forall \epsilon > 0$$

2 Strong Consistency

Let $\hat{\theta}_n$ be an estimator of θ based on sample size n :

$$\hat{\theta}_n \xrightarrow{a.s.} \theta \quad \text{as } n \rightarrow \infty$$

Meaning:

$$P\left(\lim_{n \rightarrow \infty} \hat{\theta}_n = \theta\right) = 1$$

Proof 1: Sample Mean is Consistent

Let $X_1, X_2, \dots, X_n$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$.

Sample mean:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

By the Weak Law of Large Numbers:

$$\bar{X}_n \xrightarrow{p} \mu$$

- Hence, $\bar{X}_n$ is a consistent estimator of μ .

Proof 2: Sample Variance is Consistent

Estimator:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

- $E[s^2] = \sigma^2$ (unbiasedness)
- $\text{Var}(s^2) \rightarrow 0$ as $n \rightarrow \infty$

By Chebyshev's Inequality:

$$P(|s^2 - \sigma^2| > \epsilon) \rightarrow 0$$

- Thus, s^2 is consistent for σ^2 .

Sufficient Condition for Consistency

Theorem: If $\hat{\theta}_n$ satisfies:

① $E[\hat{\theta}_n] \rightarrow \theta$ (asymptotic unbiasedness)

② $\text{Var}(\hat{\theta}_n) \rightarrow 0$

then $\hat{\theta}_n$ is consistent for θ .

Proof Sketch:

By Chebyshev's inequality:

$$P(|\hat{\theta}_n - \theta| > \epsilon) \leq \frac{\text{Var}(\hat{\theta}_n)}{\epsilon^2}$$

As $n \rightarrow \infty$, $\text{RHS} \rightarrow 0$.

Hence $\hat{\theta}_n \xrightarrow{P} \theta$.

Example of an Inconsistent Estimator

Variance estimator:

$$\tilde{\sigma}^2 = \frac{1}{n+5} \sum_{i=1}^n (X_i - \bar{X})^2$$

Expectation:

$$E[\tilde{\sigma}^2] = \frac{n-1}{n+5} \sigma^2 \rightarrow \sigma^2$$

But variance does not vanish fast enough for small n .
This estimator may be biased and inconsistent in practice.

Key Insights

- Consistency is about **large-sample behavior**.
- Unbiasedness $\nRightarrow$ Consistency.
- Consistency $\nRightarrow$ Unbiasedness.
 - Example: MLE variance is biased but consistent.
- Consistency ensures reliability of estimators as $n \rightarrow \infty$.

Summary

- Consistency means $\hat{\theta}_n \xrightarrow{P} \theta$ as $n \rightarrow \infty$.
- Sample mean and sample variance are consistent.
- Sufficient condition: asymptotic unbiasedness + variance $\rightarrow 0$.
- More important than unbiasedness for **large-sample inference**.

Property No 3: Efficiency

Multiple unbiased estimators may exist for the same parameter. We compare them by **efficiency**, which reflects variance performance. The most efficient estimator has the **smallest variance**.

Definition of Efficiency

- Let $\hat{\theta}_1$ and $\hat{\theta}_2$ be unbiased estimators of θ .
- **Relative Efficiency:**

$$e(\hat{\theta}_1, \hat{\theta}_2) = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}$$

- If $e(\hat{\theta}_1, \hat{\theta}_2) > 1$, then $\hat{\theta}_1$ is more efficient.
- The **Minimum Variance Unbiased Estimator (MVUE)** is the most efficient unbiased estimator.

Cramér–Rao Lower Bound (CRLB)

Theorem (Cramér–Rao Inequality):

For any unbiased estimator $\hat{\theta}$ of θ :

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}$$

- where Fisher Information is:

$$I(\theta) = E \left[\left(\frac{\partial}{\partial \theta} \ln f(X; \theta) \right)^2 \right]$$

- Equality holds for **efficient estimators**.
- Efficiency relative to CRLB:

$$\text{Efficiency}(\hat{\theta}) = \frac{1/I(\theta)}{\text{Var}(\hat{\theta})}$$

Proof Sketch of CRLB

- 1 Define **score function**:

$$U(\theta) = \frac{\partial}{\partial \theta} \ln f(X; \theta), \quad E[U(\theta)] = 0$$

- 2 Covariance:

$$\text{Cov}(\hat{\theta}, U(\theta)) = 1$$

- 3 Apply **Cauchy–Schwarz inequality**:

$$\text{Var}(\hat{\theta}) \cdot \text{Var}(U(\theta)) \geq [\text{Cov}(\hat{\theta}, U(\theta))]^2$$

- 4 Note $\text{Var}(U(\theta)) = I(\theta)$.

Thus:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}$$

Example 1: Normal Mean

Suppose $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known.

- Estimator: $\bar{X} = \frac{1}{n} \sum X_i$
- Variance: $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$
- Fisher Information: $I(\mu) = \frac{n}{\sigma^2}$
- CRLB: $1/I(\mu) = \frac{\sigma^2}{n}$
- $\bar{X}$ achieves CRLB $\implies$ efficient.

Example 2: Bernoulli Parameter

Suppose $X_1, \dots, X_n \sim \text{Bernoulli}(p)$.

- Estimator: $\hat{p} = \bar{X}$
- Variance: $\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$
- Fisher Information: $I(p) = \frac{n}{p(1-p)}$
- CRLB: $1/I(p) = \frac{p(1-p)}{n}$
- $\hat{p}$ achieves CRLB $\implies$ efficient.

Key Insights

- Efficiency compares estimators by variance.
- CRLB provides a benchmark: minimum achievable variance.
- Estimators achieving CRLB are **efficient**.
- Sample mean (Normal) and sample proportion (Bernoulli) are efficient.

Summary

- Efficiency measures relative performance of unbiased estimators.
- MVUE = most efficient unbiased estimator.
- Cramér–Rao bound gives minimum variance.
- Estimators that achieve CRLB are called **efficient**.

Property No 4: Sufficiency

In estimation theory, we often want to summarize the sample without losing information about the parameter.

A **sufficient statistic** captures all the information in the sample about the parameter.

Definition of Sufficiency

Let $X_1, X_2, \dots, X_n$ be a random sample with parameter θ .

A statistic $T(X_1, \dots, X_n)$ is **sufficient** for θ if the conditional distribution of the sample given T does not depend on θ :

$$f(x_1, \dots, x_n \mid T = t) \text{ does not depend on } \theta$$

Factorization Theorem

Neyman–Fisher Factorization Criterion

$T(X)$ is sufficient for θ iff:

$$f(x_1, \dots, x_n \mid \theta) = g(T(x), \theta) h(x)$$

- $g(T(x), \theta)$ depends on the data only through $T(x)$ and θ .
- $h(x)$ does not depend on θ .
- This is the main tool for finding sufficient statistics.

Proof Idea of Factorization Theorem

- **If part:** If factorization holds, then the conditional distribution given T does not involve θ , so T is sufficient.
- **Only if part:** If T is sufficient, likelihood can be decomposed into a function of T and θ , and a function free of θ .

Thus, factorization is necessary and sufficient.

Example 1: Normal Distribution

Suppose $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, with μ known, σ^2 unknown.

Joint density:

$$f(x \mid \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\sum (x_i - \mu)^2}{2\sigma^2}\right)$$

Factorization:

$$g\left(\sum (x_i - \mu)^2, \sigma^2\right) \cdot h(x)$$

- Sufficient statistic:

$$T(X) = \sum_{i=1}^n (X_i - \mu)^2$$

Example 2: Bernoulli Distribution

Suppose $X_1, \dots, X_n \sim \text{Bernoulli}(p)$.

Joint pmf:

$$f(x \mid p) = p^{\sum x_i} (1 - p)^{n - \sum x_i}$$

Factorization:

$$g\left(\sum x_i, p\right) \cdot h(x)$$

- Sufficient statistic:

$$T(X) = \sum_{i=1}^n X_i$$

Minimal Sufficiency

- A sufficient statistic is **minimal sufficient** if it is a function of every other sufficient statistic.
- Intuitively: the “smallest” sufficient statistic still carrying all information about θ .

Example: In $\text{Bernoulli}(p)$, $T(X) = \sum X_i$ is minimal sufficient.

Why Sufficiency Matters

- Reduces **data dimension**: from full sample to a summary.
- Provides foundation for **efficient estimation** (e.g. MVUEs).
- Plays a central role in **likelihood-based inference**.

Summary

- A statistic is sufficient if it captures all information about θ .
- Defined via conditional distribution independence.
- **Factorization Theorem** is the main criterion.
- Examples:
 - Normal(μ known): $T = \sum (X_i - \mu)^2$
 - Bernoulli(p): $T = \sum X_i$
- Minimal sufficiency gives the most compact summary.

Interval Estimation

- Point estimators provide a single value for an unknown parameter. But they do not reflect the **uncertainty** of estimation.
- An **interval estimator** gives a **range of plausible values** with a specified confidence level.

Definition of Interval Estimator

An interval estimator for θ is a pair $(L(X), U(X))$ such that:

$$P(L(X) \leq \theta \leq U(X)) = 1 - \alpha$$

- $1 - \alpha$: **confidence level** (e.g., 95%, 99%)
- $[L(X), U(X)]$: **confidence interval (CI)** for θ

Confidence Coefficient

- The probability $1 - \alpha$ is the **confidence coefficient**.
- Interpretation: If repeated sampling were done, $100(1 - \alpha)\%$ of the resulting intervals would contain θ .

Method of Construction

General steps:

- 1 Find a **pivot quantity**: statistic with distribution independent of θ .
- 2 Use its distribution to derive probability bounds.
- 3 Solve inequality for θ to obtain CI.

Example 1: Normal Mean, Variance Known

- If $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, σ^2 known:

Pivot:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

CI:

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- Confidence interval for μ .

Example 2: Normal Mean, Variance Unknown

- If σ^2 is unknown:

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

CI:

$$\bar{X} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$$

- Uses Student's t distribution.

Example 3: Bernoulli Proportion

- If $X_1, \dots, X_n \sim \text{Bernoulli}(p)$:

Estimator: $\hat{p} = \bar{X}$

Approximate CI (CLT):

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

- Confidence interval for p .

Properties of Interval Estimators

- ① **Confidence Level:** Probability the interval contains θ .
- ② **Precision:** Narrower intervals = more precise.
- ③ **Trade-off:** Higher confidence $\rightarrow$ wider interval.

Interpretation Caution

- A 95% CI does **not** mean θ has a 95% probability of lying in the computed interval.
- Instead: In repeated sampling, 95% of such intervals would contain θ .

Key Insights

- Interval estimators quantify **uncertainty**.
- Built using **pivot quantities**.
- Classic cases: Normal mean (known/unknown variance), proportion.
- Balance between **confidence** and **precision**.

Summary

- Interval estimators give $[L(X), U(X)]$ with confidence $1 - \alpha$.
- Constructed using pivot statistics.
- Normal mean and proportion are standard examples.
- Higher confidence $\rightarrow$ wider intervals.